



Quantum Algorithms Deep Dive

Four Modules: How Each Quantum Algorithm Races Its Classical Rival

QUANTUM ADVANTAGE LAB — ALGORITHM REFERENCE



Module 1: Grover's Search

Quantum amplitude amplification vs. classical brute-force search

The Problem

Find a marked item in an unstructured database of N elements. Classically, this requires checking items one by one — on average $N/2$ queries. Grover's algorithm finds it in $\sim\sqrt{N}$ queries, a provably optimal quadratic speedup.

How Grover's Algorithm Works

1. **Initialization:** Prepare a uniform superposition over all $N = 2^n$ states using Hadamard gates on n qubits.
2. **Oracle (O):** A unitary that flips the phase of the marked state: $|x\rangle \rightarrow -|x\rangle$ if x is the solution, $|x\rangle$ otherwise. On IonQ, the multi-controlled-Z at the oracle's core compiles to a single layer of MS gates thanks to all-to-all connectivity.
3. **Diffusion (D):** Reflects amplitudes about their mean, boosting the marked state's amplitude. Implemented as $H^{\otimes n} \cdot (\text{conditional phase flip}) \cdot H^{\otimes n}$.
4. **Iterate:** Repeat $O \rightarrow D$ approximately $\pi\sqrt{N}/4$ times.
5. **Measure:** The marked state is observed with high probability ($>99\%$ for single marked item).

QUANTUM — GROVER'S

Queries: $O(\sqrt{N})$

10-qubit (1024 items): ~25 oracle calls

20-qubit (1M items): ~804 oracle calls

IonQ advantage: Oracle is a single MS-gate layer (no SWAP routing)

CLASSICAL — LINEAR SEARCH

Queries: $O(N)$, avg $N/2$

10-qubit (1024 items): ~512 comparisons

20-qubit (1M items): ~500,000

comparisons

Scales linearly — quadratically worse than quantum

What the Visualization Shows

A **search-space heatmap**: each cell represents one database element. Grover's panel shows probability amplitude building on the target (glowing brighter each iteration) while surrounding states dim. The classical panel highlights cells one-by-one in sequential scan order. Users see the quantum solver "zoom in" on the answer while classical plods linearly.



Module 2: Quantum Walks on Graphs

Quantum walk spreading vs. classical random walk diffusion

The Problem

Explore a graph to find a marked vertex or measure spreading behavior. Classical random walks diffuse slowly (standard deviation grows as $\sqrt{t}$). Quantum walks spread ballistically (standard deviation grows as t), offering quadratic speedup for spatial search and hitting-time problems.

How Discrete-Time Quantum Walks Work

1. **State space:** A walker on a graph with d -regular vertices. The Hilbert space is the tensor product of position (vertex) and coin (direction) registers.
2. **Coin operator (C):** A unitary on the coin register — typically a Grover diffusion coin for symmetric graphs. Creates superposition over all outgoing edges.
3. **Shift operator (S):** Moves the walker along each edge in superposition: $|v, d\rangle \rightarrow |\text{neighbor}(v,d), d'\rangle$. On IonQ, the all-to-all connectivity implements the shift operator without SWAP chains even for densely connected graphs.
4. **Iterate:** Apply $C \rightarrow S$ for t steps. Interference causes the quantum walker's probability to spread ballistically.
5. **Measure:** Position register yields the walker's location.

QUANTUM — QUANTUM WALK

Spreading: Ballistic, $\sigma \sim t$

Hitting time: $O(\sqrt{N})$ on many graph families

Behavior: Coherent interference creates non-classical probability distributions

IonQ advantage: Dense shift operators stay shallow

CLASSICAL — RANDOM WALK

Spreading: Diffusive, $\sigma \sim \sqrt{t}$

Hitting time: $O(N)$ on many graph families

Behavior: Gaussian diffusion, no interference

Memoryless: Each step independent of history

What the Visualization Shows

A **force-directed graph** with animated walkers. The quantum panel shows probability amplitude flowing outward in a wave-like pattern with visible interference fringes (some paths reinforce, others cancel). The classical panel shows a diffusing cloud that spreads slowly and uniformly. The contrast between ballistic and diffusive spreading is striking and immediate.



Module 3: Variational Quantum Eigensolver (VQE)

Hybrid quantum-classical optimization vs. purely classical solvers

The Problem

Find the ground-state energy of a molecular Hamiltonian H . This is central to computational chemistry — predicting molecular properties, reaction energies, and material behavior. Exact classical diagonalization scales as $O(2^n)$, becoming intractable beyond ~ 50 spin-orbitals.

How VQE Works

- Ansatz preparation:** A parameterized quantum circuit $U(\theta)$ prepares a trial state $|\psi(\theta)\rangle$. We use a **hardware-efficient ansatz** (HEA) with layers of single-qubit rotations and MS entangling gates, designed for IonQ's all-to-all connectivity — every qubit pair can entangle directly in each layer.
- Energy measurement:** Decompose H into a sum of Pauli strings: $H = \sum_i c_i P_i$. Measure $\langle \psi(\theta) | P_i | \psi(\theta) \rangle$ for each term on IonQ hardware.
- Classical optimization:** A classical optimizer (COBYLA, L-BFGS-B, or SPSA) updates θ to minimize $E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle$.
- Iterate:** Repeat steps 1–3 until energy converges to the ground state.

IonQ-specific design: The HEA exploits full connectivity: each entangling layer applies MS gates between all qubit pairs (or a dense subset), creating highly expressive ansätze impossible on nearest-neighbor hardware without massive SWAP overhead. Fewer layers $\rightarrow$ fewer parameters $\rightarrow$ faster convergence $\rightarrow$ less noise accumulation.

QUANTUM — VQE ON IONQ

State space: Exponential (2^n) encoded in n qubits

Expressivity: Full-connectivity ansatz in $O(1)$ MS layers

Target molecules: H_2 , LiH , BeH_2 (6–14 qubits)

Advantage: Explores exponential Hilbert space natively

CLASSICAL — GRADIENT OPTIMIZATION

Method: Hartree-Fock + CCSD(T) or exact diag.

Scaling: $O(N^7)$ for CCSD(T), $O(2^N)$ for exact

Limitation: Exponential wall at ~ 50 orbitals

Shown as: Classical energy converging via SciPy

What the Visualization Shows

An **energy-landscape contour plot**: the x/y axes represent two principal variational parameters (projected from higher-D), the z -axis (color) shows energy. The quantum optimizer's trajectory is plotted as it descends toward the minimum, with each point

representing a real IonQ measurement. The classical solver traces its own convergence path. Users see both solvers racing toward the same ground-state energy from different starting points.



Module 4: Hamiltonian Simulation

Native quantum time-evolution vs. classical matrix exponentiation

The Problem

Simulate the time-evolution of a quantum system: given Hamiltonian H and initial state $|\psi_0\rangle$, compute $|\psi(t)\rangle = e^{-iHt}|\psi_0\rangle$. Feynman's original motivation for quantum computers — "nature isn't classical, so use a quantum computer to simulate it." Classically, storing the state requires 2^n complex amplitudes and exponentiating a $2^n \times 2^n$ matrix.

How Trotterized Simulation Works

- Hamiltonian decomposition:** Write $H = \sum_k H_k$ where each H_k is easy to exponentiate (e.g., individual Pauli terms or pairs).
- Trotter-Suzuki formula:** Approximate $e^{-iHt} \approx (e^{-iH_1\Delta t} \cdot e^{-iH_2\Delta t} \cdot \dots \cdot e^{-iH_k\Delta t})^r$ for r Trotter steps with $\Delta t = t/r$. Error scales as $O(t^2/r)$.
- Circuit compilation:** Each $e^{-iH_k\Delta t}$ maps to a short sequence of native gates. For Pauli-pair terms (e.g., XX , ZZ), the IonQ MS gate implements the interaction directly — it is natively an XX -rotation.
- Execute and measure:** Run the Trotterized circuit on IonQ, measure observables at each time step.

IonQ's native advantage for simulation: The Mølmer-Sørensen gate is physically an XX interaction — exactly the kind of term that appears in spin-chain and molecular Hamiltonians. Where other platforms must decompose XX interactions into multiple CNOT + rotation sequences, IonQ implements them in a single native gate. For a Heisenberg spin chain with N sites, this saves $\sim 3\times$ gates per Trotter step.

QUANTUM — TROTTERIZED ON IONQ

Resources: n qubits for n -site system

Gate cost per step: $O(n^2)$ MS gates (all-pairs)

MS gate = native XX : No decomposition needed

Scaling: Polynomial in n and t

CLASSICAL — MATRIX EXPONENTIATION

Resources: $2^n \times 2^n$ dense matrix

Cost per step: $O(2^{2n})$ for dense, $O(2^n)$ for sparse

Memory wall: ~ 30 qubits = 8 GB just for the state vector

Scaling: Exponential in n

What the Visualization Shows

Matrix-norm decay curves: the y-axis shows the fidelity (overlap) between the simulated state and the exact solution over time. The quantum curve shows IonQ's Trotterized evolution tracking the exact solution closely (bounded Trotter error), while the classical

panel shows computation time growing visibly as the system size increases. A secondary panel displays **spin-chain magnetization dynamics** — an animated 1D lattice showing spin orientations evolving in real time, quantum vs. classical side by side.

Algorithm Comparison Summary

MODULE	QUANTUM ALGORITHM	CLASSICAL BASELINE	QUANTUM SPEEDUP	IONQ-SPECIFIC BENEFIT
Search	Grover's	Linear scan	Quadratic: $O(\sqrt{N})$ vs $O(N)$	Oracle = 1 MS-gate layer (no SWAP routing)
Walks	Discrete quantum walk	Random walk	Quadratic: ballistic vs diffusive	Dense shift operator stays shallow
VQE	Variational eigensolver	CCSD(T) / exact diag.	Exponential (state space)	Full-connectivity ansatz, fewer layers
Simulation	Trotterized evolution	Matrix exponentiation	Exponential (memory + time)	MS gate = native XX interaction

Qubit and Circuit Requirements

MODULE	QUBITS	CIRCUIT DEPTH (IONQ)	CIRCUIT DEPTH (GRID TOPOLOGY)	DEPTH REDUCTION
Grover's (10-qubit)	10	~50 layers	~200 layers	~4x
Quantum Walk (12-qubit)	12	~30 layers / step	~90 layers / step	~3x
VQE - LiH (10-qubit)	10	~20 layers / iteration	~80 layers / iteration	~4x
Simulation - Heisenberg (16-qubit)	16	~40 layers / step	~120 layers / step	~3x

Key takeaway: IonQ's all-to-all connectivity reduces circuit depth by 3-4x compared to grid-topology devices across all four modules. Shallower circuits mean less decoherence, higher fidelity outputs, and cleaner visualizations — directly enabling the educational mission of Quantum Advantage Lab.